

EV PULSE NC: DATA-DRIVEN EV INFRASTRUCTURE GAP ANALYSIS FOR NORTH CAROLINA

ABSTRACT

North Carolina holds \$109 million in federal NEVI Formula Program funding for public EV charging infrastructure but lacks a data-driven method for prioritizing county-level allocation against utilization efficiency and Justice40 equity, amid BEV adoption growing at a 53.8% compound annual rate. This study integrates five analytical phases into a two-tier NEVI Priority Score combining a county-level ranking with ZIP-level targeting: out-of-sample forecast validation, an updated NREL AFDC infrastructure baseline, ZIP-level density analysis with an additive Theil-T decomposition, LEHD LODS workplace-charging demand modeling, and a CEJST v2.0 Justice40 overlay. Forecast validation yielded a mean absolute percentage error of 4.34%; the Theil-T decomposition found that 84.5% of charging-infrastructure inequality occurs within counties rather than between them, motivating the two-tier design. The three scoring pillars proved statistically independent (maximum VIF 1.41) with stable top-three rankings (Union, Mecklenburg, Guilford) under sensitivity perturbation. The framework informs state transportation-agency allocation decisions, captures 73% of the statewide BEV fleet, and is fully reproducible from public data.

1. INTRODUCTION

1.1 Overview

EV Pulse NC analyzes alignment between EV adoption and public charging infrastructure across North Carolina's 100 counties. Five analytical phases (Phases 1 through 5, defined in Section 4.1) converge in a two-tier scoring framework (a county-level NEVI Priority Score and a ZIP-level equity targeting layer) ranking the top 10 NC counties by BEV registration count for federal NEVI Formula Program funding.

1.2 Motivation

North Carolina's battery electric vehicle (BEV) fleet grew from 5,165 registered vehicles in September 2018 to 94,371 by June 2025, a 1,727% increase and 53.8% compound annual growth rate. Growth is uneven: Wake County alone holds more BEV registrations than the bottom 60 counties combined, and the county-level Gini coefficient reaches 0.805. As of January 2026, the state has 1,985 stations housing 6,145 connectors (roughly 16.9 BEVs per public port, exceeding the recommended 10 to 15 benchmark), and several BEV-holding counties have zero public stations. The federal NEVI Formula Program allocated \$109 million. A demand-driven allocation reinforces existing concentration; an equity-driven one places stations where utilization is too low to sustain. In February 2026, NCDOT shifted from a corridor-only approach (50 interstate stations) toward 16 rural and community locations; the top 10 counties hold 72% of all BEVs while the bottom 50 hold just 4.8%.

1.3 Goals

Five questions drive the work. First, how accurate are SAS Model Studio's county-level BEV forecasts against unseen data? Validation achieved a MAPE of 4.34% across all 100 counties, with 69.00% systematic underprediction signaling accelerating adoption. Second, where are the largest demand-supply gaps? A full NREL AFDC API pull replaced the DCFC-only July 2024 extract, capturing all charging levels. Third, how do ZIP-level density patterns reveal priorities county aggregation obscures (Wake County spans 857 square miles). Fourth, how do commuting flows reshape demand once workplace needs are layered onto residential data, which captures roughly 80% of charging behavior? Fifth, can a weighted scoring equation translate these layers into defensible recommendations, via $NEVI\ Score = w1*Equity + w2*Utilization + w3*Cost_Effectiveness$?

1.4 Project Background

EV Pulse NC is an independent, self-directed project using only public data. Its baseline processed 8,200 county-month observations across 100 counties and 82 months of NCDOT registration data, paired with 1,725 connectors across 355 AFDC stations, establishing ARIMA forecasts with a weighted in-sample MAPE of 2.73% and the 0.805 Gini concentration. But county aggregation masked intra-county variation, the snapshot was 18 months old, forecasts were untested out-of-sample, and every BEV was treated as residential. The five-phase extension plus scoring layer addresses these. Phase 1 validated forecasts against four months of held-out data (MAPE 4.34%); Phase 2 upgraded supply to a February 2026 AFDC API pull (all charging levels); Phase 3 measured ZIP-level inequality via Gini and additive Theil-T decomposition. Phase 4 added workplace demand from LEHD LODES origin-destination flows; Phase 5 layered a Justice40 equity overlay using the Climate and Economic Justice Screening Tool (CEJST v2.0). ZIP-level BEV registration is unavailable from NCDOT, and LODES 2021 data predates fuller post-COVID return-to-office, so the analysis reports ranges where data is weak.

2. PROBLEM DEFINITION

2.1 Statement of the Problem

North Carolina holds \$109 million in federal NEVI Formula Program funding but no publicly available, data-driven method for allocating it across counties. Its BEV fleet has grown at a 53.8% compound annual rate since 2018, but so unevenly that the county-level Gini coefficient reaches 0.805. Infrastructure is both insufficient statewide (16.9 BEVs per port versus a 10-to-15 benchmark) and distributed to reflect existing demand, not projected need, so efficiency and equity pull in opposite directions. NCDOT's February 2026 pivot from a corridor-only NEVI strategy to one including 16 rural and community locations confirms the tension; this project supplies a systematic method to resolve it.

2.2 Scope and Boundaries

The analysis covers all 100 North Carolina counties. Demand spans September 2018 through October 2025: 86 months of NCDOT BEV registration data, 8,600 county-month observations (8,200 baseline plus 400 from the four-month validation holdout). Supply comes from the NREL AFDC API as of February 2026: 6,145 connectors across 1,985 stations, all charging levels and access types, replacing the baseline's DCFC-only extract (1,725 connectors across 355 stations, July 2024), which missed the 82.4% of stations offering only Level 2 charging. Three scales structure the work: county-level, the primary unit for NEVI allocation; ZIP-code-level, covering 134 urban ZIP codes within the top 10 counties (73% of the statewide BEV fleet); and Census tract-level via the CEJST overlay mapping Justice40 disadvantaged-community boundaries. Temporal windows differ by source, with SAS Model Studio forecasts validated against July through October 2025 actuals and LEHD LODES 2021 origin-destination flows adjusted for post-COVID remote work via a uniform multiplier.

2.3 Analytical Complexity

A county's NEVI priority spans four dimensions no single metric captures: the size of its demand-supply gap, its growth rate, whether it contains disadvantaged communities satisfying the Justice40 mandate (40% of benefits directed to underserved populations), and whether workplace commuting creates demand invisible to residential registration data. The scoring equation, $\text{NEVI Score} = 0.40 \times \text{Equity} + 0.35 \times \text{Utilization} + 0.25 \times \text{Cost Effectiveness}$, compresses these into one ranked output with non-arbitrary weights. Equity receives the highest weight (0.40), an analytical anchor reflecting the Justice40 40%-of-benefits target established under Executive Order 14008 (January 2021). It is informed by, not mandated by, current policy, since EO 14008 was rescinded in January 2025 and, while the IJA-specific NEVI rule and federally-approved state deployment plans retain meaningful-public-involvement

and Justice40-aligned considerations, the explicit 40%-benefit-flow target no longer carries Executive Order standing. Utilization receives 0.35 because validated BEV forecasts are the project's strongest data. Cost-effectiveness receives the lowest weight (0.25) because workplace charging estimates are the most uncertain, relying on 2021 LODES commuting data that predates fuller post-COVID return-to-office and may overstate flows by 10 to 25%. Component scores travel with the composite, so nothing is hidden.

2.4 Success Criteria

The hardest bar is practical defensibility: the top-ranked counties should make intuitive sense to anyone familiar with North Carolina's EV environment, and where they do not, the components should explain why, with the methodology resting on validated forecasts, current infrastructure data, and federal equity mandates. The deliverable is a ranked list of the top 10 NC counties with composite NEVI priority scores, component-level breakdowns, and a framework extensible to the remaining 90 counties. An optional external check compares the rankings against NCDOT's February 2026 deployment decisions: agreement with its planned 16 rural/community locations would suggest the methodology captures real policy priorities, while disagreement would reveal where the weights diverge from state preferences.

2.5 Exclusions

The project does not address site-level placement: flagging that ZIP code 27603 in Raleigh needs charging is in scope, but selecting the specific parking lot is not. Private and residential charging, roughly 80% of all EV charging, falls outside it because NEVI funds public infrastructure. Station-level real-time utilization data is unavailable, so BEV-to-port ratios proxy demand pressure rather than measured throughput. The boundary is North Carolina; cross-state commuting from Virginia and South Carolina is not modeled, though it likely affects border counties. The project also does not model NEVI's political dynamics; the equation optimizes for equity, utilization, and cost-effectiveness, while real decisions involve legislative priorities and stakeholder negotiations beyond its reach.

3. LITERATURE REVIEW

3.1 EV Adoption Trends and Geographic Concentration

EV adoption growth is well documented, its spatial distribution less so [17]; California alone holds roughly 40% of national registrations, and concentration repeats within states. North Carolina grew at a 53.8% compound annual rate between 2018 and 2025, yet Wake County holds more registrations than the bottom 60 counties combined (county Gini 0.805; [12]). As sales outpace deployment [9], North Carolina's 16.9 BEVs per port exceeds the DOE's recommended 10-to-15 benchmark, and the statewide figure obscures county-level variation that motivates a sub-county approach.

3.2 Infrastructure Gap Measurement

The demand-supply mismatch is often framed as facility location, via the p-median and maximal coverage problems of Hakimi [7] and ReVelle and Swain [13]; Celik and Ok [3] added discrete-event simulation, but such models assume fixed demand. NC work addressed pieces: a Raleigh gas-station siting study [20], the GreenEVT traffic-grid testbed [11], and a Cary utilization study [10]. None of these addresses distributional equity. This project instead treats distribution as an inequality problem; the BEV-to-port ratio is the primary county- and ZIP-level metric, but ratios cannot capture within-unit evenness.

3.3 Inequality Measurement: Gini and Theil Decomposition

The Gini coefficient is the standard inequality measure, but it cannot separate between-county from within-county inequality. The Theil-T (GE(1)) measure of Theil [15], formalized for subgroup

decomposition by Shorrocks [14], decomposes exactly without residual. Prior EV studies used Theil as a scalar: Choi, Xu, and Jiao [5] reported a single magnitude for Austin, Texas. What has not been done is exploiting the additive decomposability of the GE(1) class [2, 14] to partition charging inequality into between- and within-county components. A scalar reports *magnitude*; a decomposition reports *structure*. This project found 84.5% of inequality in the top 10 NC counties is within-county. County-level formulas, how most state NEVI programs distribute funding, would miss the majority, justifying a two-tier approach: county rankings for where to invest, ZIP-level targeting for whom.

3.4 Justice40 and the CEJST Policy Context

The Justice40 Initiative, established by Executive Order 14008 in January 2021, targeted that 40% of certain federal benefits flow to disadvantaged communities; the NEVI Formula Program (IIJA Section 11401; codified at 23 U.S.C. §175) fell under it. EO 14008 was rescinded in January 2025, removing the explicit target, though the NEVI rule and FHWA-approved state plans retain Justice40-aligned considerations; this study treats the 0.40 equity weight as an analytical anchor, not a mandate. The operational definition comes from CEJST v2.0 [6], flagging tracts both low-income (at or above the 65th percentile) and overburdened on at least one of eight environmental, health, or infrastructure dimensions. In North Carolina, 43.0% of evaluable tracts are disadvantaged; for the top 10 BEV counties the rate is 18.5%, but nearly one in five residents still lives in a disadvantaged tract. Federal programs increasingly use multi-criteria scoring (USDOT RAISE, BIL CFI); this study's contribution is its integration of an additive Theil-T decomposition with the scoring architecture (Section 3.3).

3.5 Workplace Charging and Commuting Demand

Most research treats demand as residential, capturing roughly 80% of behavior but missing employment centers, whose commuter demand is invisible to registration data. LEHD LODES provides block-level flows: Mecklenburg absorbs +194,361 commuters per day [16], who park 8 to 10 hours, a fit for Level 2. The workplace literature is thin. Uimonen and Lehtonen [18] modeled office load profiles, and the UCLA Luskin DCF model [4] found workplace installations reach positive NPV faster, but none identified estimates demand at scale from commuter flows for North Carolina. This project filters LEHD LODES 2021 through SE03 segmentation (workers above \$40,000), an ACS correction for households above \$75,000 (the EV affordability threshold), and a 0.85 remote-work multiplier [1], yielding 859,260 statewide for the cost-effectiveness pillar. The 15% remote-work reduction is a national average, so results are reported at baseline (30% charging adoption rate) and low/high scenarios.

3.6 Scoring and Prioritization Precedents

Multi-criteria prioritization is established in transportation planning. NREL projected national investment needs by region [19] without county-level scores, and the DOE AFDC offers station-level rather than geographic-allocation tools. Less common is combining equity with utilization and cost-effectiveness into one geographic-prioritization equation. This project's NEVI Score = $0.40 \times \text{Equity} + 0.35 \times \text{Utilization} + 0.25 \times \text{Cost-Effectiveness}$ takes that step, weights reflecting data confidence (Section 2.3). Weight sensitivity tested equity weights from 0.30 to 0.50 in 0.05 increments, the first of three robustness checks (Section 4.7.1); the top three counties (Union, Mecklenburg, Guilford) held across all five scenarios.

3.7 Synthesis and Gap

The literature provides mature siting tools (facility location), scalar inequality measures (Gini, Theil as in [5]), a federal equity framework (Justice40 / CEJST, with caveats in Section 3.4), and recognition that workplace commuting creates demand invisible to residential data. Missing is a study integrating all of these into one pipeline for a specific state's NEVI allocation. This study's contribution is twofold. First, it applies the additive decomposability of the GE(1) class [2, 14] to EV charging infrastructure inequality, a

structural decomposition that prior scalar work [5] does not perform. Second, it integrates that decomposition into a two-tier framework combining a county-level NEVI Priority Score with ZIP-level equity targeting. This combination of residential demand, workplace flows, federal equity designations, and validated forecasts in a weighted equation has not been published for North Carolina or, as far as this review can determine, any other state.

4. METHODOLOGY

4.1 Analytical Pipeline

The study executes five analytical phases that integrate into a single prescriptive scoring framework. Each phase has defined inputs, outputs, and validation evidence:

- **Phase 1: Forecast validation.** Out-of-sample testing of SAS Model Studio BEV adoption forecasts against July through October 2025 actuals (Section 6.1).
- **Phase 2: Infrastructure baseline.** Upgrade from a DCFC-only July 2024 extract (355 stations, 1,725 connectors) to a full February 2026 AFDC API pull (1,985 stations, 6,145 connectors) (Section 6.2).
- **Phase 3: ZIP-level inequality.** Within-county infrastructure-density analysis using Gini coefficient and additive Theil-T decomposition (Sections 4.4, 6.3).
- **Phase 4: Workplace charging demand.** Origin-destination commuter flow modeling using LEHD LODS 2021 (Section 4.6).
- **Phase 5: Justice40 equity overlay.** Area-weighted tract-to-ZCTA crosswalk integrating CEJST v2.0 disadvantaged-community designations (Section 4.5).

The NEVI scoring framework integrates the outputs of all five phases into a composite Priority Score with three pillars (Equity, Utilization, Cost-Effectiveness):

$$\text{NEVI Score} = 0.40 \times \text{Equity} + 0.35 \times \text{Utilization} + 0.25 \times \text{Cost-Effectiveness}$$

The weight rationale is discussed in Section 2.3. Sensitivity analysis tested equity weights from 0.30 to 0.50 in 0.05 increments; the top three counties (Union, Mecklenburg, Guilford) remained stable across all scenarios, though their internal ordering shifted in expected ways.

The five phases are not strictly sequential. Exploratory profiling of the LEHD LODS data in Phase 4 sent the pipeline back through an ACS-based income enrichment step, and the Phase 1 validation findings (systematic underprediction, 69.00% of forecasts below actuals) propagated forward to inform how forecast-based allocations are buffered.

4.2 Demand-Side Analysis

The demand side rests on NCDOT's complete BEV registration time series. The baseline dataset comprises 8,200 county-month observations spanning 100 counties across 82 months (September 2018 through June 2025). The validation extension added 400 observations from the July-through-October 2025 holdout period, bringing the total to 8,600 county-month records.

County-level ARIMA forecasting was performed in SAS Model Studio, which auto-selected the best model type per county: exponential smoothing for 82 counties, ARIMA for 13, and unobserved component models for 5. The forecasts project BEV adoption through June 2028 with a weighted in-sample MAPE of 2.73%. Out-of-sample validation against the four-month holdout produced a MAPE of **4.34%**, below the 5% threshold the project treats as the usability cutoff.

The validation also uncovered a systematic bias. Actual registrations exceeded forecasts in **69.00%** of observations, with a mean bias of **+18.22 vehicles per county-month**. The 95% prediction interval

achieved only **75.50%** raw coverage; after county-specific bias correction, coverage recovers to **93.75%** (decomposed in §4.9). The underprediction was most pronounced in high-growth urban counties; Mecklenburg County's October 2025 actual exceeded its forecast by 975 vehicles. This pattern is consistent with an adoption curve that has steepened since the Inflation Reduction Act tax credits took effect, and it means forecast-based infrastructure planning needs an upward buffer of 4 to 5% to avoid systematic underinvestment.

A binary flag (`Methodology_PostMay2025`) marks an NCDOT registration-counting methodology change after May 2025; the underlying data structure remained identical, so the existing pipeline handled it without modification. The change's magnitude is approximately 0.4% of statewide totals (370 vehicles), contributing roughly 20% of the observed underprediction bias, and is characterized further in §4.9 and §5.1.

4.3 Supply-Side Analysis

The baseline analysis from Fall 2025 used a manually exported AFDC dataset filtered to DC fast chargers only: 1,725 connectors across 355 stations. A full programmatic pull from the NREL AFDC API replaced that extract, capturing 6,145 connectors across 1,985 stations in all charging levels (Level 1, Level 2, and DCFC) and all access types (1,876 public, 109 private). All subsequent analysis uses the February 2026 API pull exclusively.

The DCFC-only extract captured nearly all fast-charging ports (1,742 of 1,745 statewide) but missed 4,298 Level 2 ports and all 35 Level 1 ports. Since Level 2 chargers account for 82.4% of all stations and are the natural fit for workplace charging (4 to 8 hours at 7 kW matches a typical workday), any gap analysis built on DCFC-only data would systematically misrepresent supply in exactly the counties where workplace charging demand is highest.

Supply-side analysis operates at the station level, with port counts stored in aggregate columns per charging level. A station with two 7 kW Level 2 chargers delivers fundamentally different capacity than one with eight 250 kW DC fast chargers; the connector-level granularity matters for capacity-weighted BEV-to-port ratios and for cost-effectiveness estimates that depend on throughput potential.

4.4 Inequality Measurement

Infrastructure inequality is measured through two complementary indices applied at the ZIP code level within each county. The Gini coefficient captures the overall shape of the distribution: a county-level Gini of 0.623 for Mecklenburg shows that charging access varies enormously by ZIP code. The statewide population-weighted Gini of 0.566 sits below the 0.805 Gini for BEV ownership concentration, suggesting infrastructure is partially tracking demand but not equitably.

The Gini cannot decompose inequality into its spatial components, which is where the Theil index enters. The Theil-T ($GE(1)$) measure, following the decomposition methodology formalized by Shorrocks [14] and the additive-decomposability framework of Bourguignon [2], partitions total inequality into between-county and within-county components with no residual. The decomposition verified to machine precision (difference of $2.22e-16$ between summed components and the total). A robustness check using Theil-L ($GE(0)$) confirmed the pattern.

The result was unambiguous: 84.5% of charging infrastructure inequality in the top 10 NC counties occurs within counties, not between them. The policy implication is discussed in Section 3.3; county-level allocation formulas, which is how most state NEVI programs currently distribute funding, would miss the majority of the inequality problem.

The Theil decomposition is a diagnostic input to architectural choice, not a parameter input to either tier's score. Neither the NEVI Priority Score (Tier 1, the county-level ranking) nor the ZIP-level equity score (Tier 2, the sub-county targeting layer) takes Theil components as variables. Tier 1 uses equity, utilization, and cost-effectiveness pillars; Tier 2 uses population, disadvantaged-community status, and station density. The 84.5% within-county finding justifies the existence of Tier 2 but does not enter either tier's computation as an input.

4.5 Equity Overlay

This study's CEJST integration follows the standard established by EPA EJScreen and the HUD USPS Crosswalk for area-weighted spatial integration of federal equity data. The equity dimension integrates Justice40 disadvantaged community designations from CEJST v2.0 (December 2024 release) into the scoring pipeline. As described in Section 3.4, CEJST flags Census tracts that are both low-income and overburdened on at least one of eight environmental, health, or infrastructure dimensions. In North Carolina, 934 of 2,170 evaluable tracts (43.0%) carry the designation, substantially above the national average.

Operationalizing CEJST data required a spatial crosswalk between 2010-vintage Census tracts (the boundaries CEJST uses) and 2020-vintage ZCTAs (the sub-county unit for infrastructure analysis). The crosswalk uses area-weighted interpolation in the NC State Plane projection (EPSG:32119), the same approach used by EPA EJScreen and HUD's USPS Crosswalk. Each ZCTA's disadvantaged population share is estimated as the area-weighted average of the binary disadvantaged flags from its constituent tract fragments. Intersection polygons smaller than 100 square meters were discarded as geometric artifacts, and border-state tracts from Virginia, South Carolina, Tennessee, and Georgia were included to ensure complete coverage of cross-border ZCTAs. Twenty-five zero-population tracts (12 water-body, 13 institutional and military) were excluded from the denominator.

The crosswalk introduces a known limitation: 2010 tract boundaries do not perfectly align with 2020 ZCTA boundaries, and area-weighted interpolation assumes uniform population distribution within tracts. At the county level this is irrelevant because North Carolina's county boundaries are geographically identical between the two vintages. At the sub-county level, the error is modest in urban areas (small tracts) but larger in rural tracts spanning heterogeneous land uses. A dasymetric refinement using population grids is noted as future work.

A sensitivity analysis tested the equity scores against CEJST's most debatable component. Removing the climate change category entirely (the only category incorporating forward-looking model projections) reduced the statewide disadvantaged rate from 43.0% to 37.3%. Seven of ten study counties showed zero impact; the three affected counties (New Hanover, Durham, Wake) shifted by 2.2 to 4.7 percentage points. The equity rankings are not sensitive to this choice.

4.6 Workplace Charging Demand

The baseline treated BEV demand as purely residential, capturing approximately 80% of charging behavior but ignoring employment centers. Phase 4 adds workplace charging demand using the Census Bureau's LEHD LODES 2021 origin-destination commuter flows at the Census block level, replacing the originally planned CTPP dataset on three criteria: more recent vintage (2021 versus 2012–2016), finer geography (block-level versus county-level), and free direct download.

The raw LODES data contains 3,768,428 intra-NC commuter flow records. Transforming these into workplace charging demand estimates requires a three-layer adjustment pipeline. The first layer filters for SE03 workers (those earning above \$40,000 annually), reducing the pool to 2,041,731. The second layer applies a county-specific ACS income correction factor, calculated as the ratio of households above

\$75,000 to those above \$40,000 in each origin tract, to approximate EV-affordable income levels and bridge the mismatch between LODES individual earnings and ACS household income. This filter reduces the count to 1,010,894. The third layer applies a 0.85 remote work multiplier from Barrero, Bloom, and Davis [1], producing a final adjusted demand estimate of 859,260 statewide.

An earlier fourth layer adjusting for renter tenure (on the theory that renter-heavy tracts have greater workplace charging need due to limited home charging) was removed during advisor review. No NC-specific calibration data exists, and the national literature on renter EV adoption rates (3 to 4 times lower than homeowner rates) would pull the multiplier in the opposite direction. Renter share data is retained for descriptive reporting and feeds into the equity pillar through CEJST's housing burden category, but does not enter the demand calculation directly.

The income correction is an approximation. LODES reports individual earnings in three coarse bins, ACS B19001 reports household income across 16 bins (a worker earning \$35,000 might live in a \$120,000 household), and the correction factor assumes the income composition of commuters matches that of their origin tract's households. The project reports results at \$60,000, \$75,000, and \$100,000 income thresholds to bracket this uncertainty.

Net commuter flow analysis classified all 100 NC counties into three typologies: 8 employment centers (net inflow above +10,000), 75 balanced counties, and 17 bedroom communities (net outflow below -10,000). Mecklenburg absorbs the largest net inflow (+194,361 commuters per day), followed by Wake (+126,517) and Durham (+89,450). The workplace port estimates from these flows feed into the cost-effectiveness pillar of the NEVI scoring equation at a baseline 30% charging adoption rate and 15:1 port ratio, with low and high scenarios bracketing the assumptions.

4.7 NEVI Scoring Construction

The scoring equation combines three pillars into a single composite ranking for each county. Each pillar is itself a weighted composite of sub-metrics, all min-max normalized to a 0-to-1 scale.

The equity pillar (weight 0.40) draws on four sub-metrics: Justice40 population-weighted disadvantaged percentage (weight 0.40 within the pillar), population-weighted Gini coefficient (0.30), count of underserved ZIP codes (0.20), and percentage of ZIP codes with zero stations (0.10). The utilization pillar (weight 0.35) draws on two sub-metrics: BEV-to-port ratio from the February 2026 AFDC infrastructure data (primary differentiator) and a forecast underprediction buffer of 0.045 applied globally. The cost-effectiveness pillar (weight 0.25) draws on three sub-metrics: workplace charging efficiency, net commuter demand magnitude, and population density.

Of the nine designed sub-metrics, seven actively differentiate counties in the 10-county study cohort. Two carry zero variance: `zero_station_pct` is 0.0 for all 10 counties because every top-10 urban county has at least some stations in every inhabited ZIP, and `forecast_buffer` is a uniform 0.045 applied globally rather than per-county. Both would contribute signal if the scoring were extended to all 100 counties, where rural counties have zero-station ZIPs and county-specific forecast errors could replace the global buffer. The effective scoring resolution for the 10-county analysis relies on three equity sub-metrics (`justice40_pct`, `gini_weighted`, `underserved_zips`), one utilization sub-metric (`bev_per_port`), and three cost-effectiveness sub-metrics (`workplace_efficiency`, `commuter_demand`, `pop_density`).

Union County's utilization score illustrates a compression problem inherent in min-max normalization. At 101.5 BEVs per port, Union is more than three times the next-highest county (Cabarrus at 32.0). Normalizing to the 0-to-1 scale makes Union 1.0 and compresses the remaining nine counties into the

lower portion of the utilization scale, reducing the pillar's ability to differentiate mid-ranked counties. The deficit is genuine, not a data artifact, but the scoring is sensitive to extreme values in small-n cohorts.

The final rankings place Union first (0.561), Mecklenburg second (0.548), and Guilford third (0.465). These three represent distinct archetypes: Union is utilization-driven (suburban growth outpacing infrastructure), Mecklenburg is equity-driven (concentrated disadvantaged tracts in Charlotte's urban core), and Guilford combines high equity burden with moderate utilization strain. Orange County ranks last (0.077), reflecting low Justice40 burden, adequate per-capita infrastructure, and a university-town demographic. Weight sensitivity analysis confirms the top-three set is stable across all tested scenarios; as equity weight rises, Mecklenburg gains and Union falls.

4.7.1 Observational, Not Experimental: OFAT Disclosure

The weight sensitivity analysis described above and in Section 6 perturbs the three pillar weights one factor at a time (OFAT) across the 0.30–0.50 equity-weight range while holding the utilization-to-cost-effectiveness ratio constant. This is computational sensitivity analysis, not a fractional-factorial or response-surface design. The "triply-robust" top-3 finding is that Union, Mecklenburg, and Guilford hold their positions across three independent perturbations: the cost-effectiveness sub-weight extremes documented in Section 4.7, the equity-weight range, and the remote-work multiplier values mathematically demonstrated to cancel in min-max normalization. This is robustness evidence under that restricted exploration, not a claim of design optimality across the full weight simplex; a full design-of-experiments treatment over the simplex is identified as future work. The analysis is observational rather than experimental: counties are not randomly assigned to treatment conditions, demographic variables are not manipulated, and policy parameters are not set by the analyst.

4.8 Tools and Environment

All analysis was performed in Python, primarily in modular scripts organized by pipeline stage. GeoPandas and Shapely handled spatial operations including the tract-to-ZCTA crosswalk, station geocoding, and choropleth generation. Pandas and NumPy supported the tabular data pipeline across all five phases. Statsmodels provided ARIMA modeling, stationarity testing, and diagnostic checks for the time series forecasting. Matplotlib produced the publication-quality figures at 600 DPI with PDF export. The demand-side forecasting used SAS Model Studio, which auto-selected model types per county and generated the predictions validated in Phase 1. The full diagnostic battery (Variance Inflation Factor, Chow test, CI coverage decomposition, estimator method comparison) is consolidated in §4.9 below.

4.9 Diagnostic Suite

This subsection consolidates four diagnostics applied to the scoring framework and the underlying forecast inputs, otherwise scattered across §4.2, §4.4, §4.8, and §6.1.

Multicollinearity (Variance Inflation Factor). A VIF check on the three scoring pillars produced maximum VIF 1.41 (Equity), with Utilization at 1.08 and Cost-Effectiveness at 1.32, all well below the conventional 5.0 concern threshold. Pairwise Pearson correlations confirm the same: Equity ↔ Utilization $r = -0.26$, Equity ↔ Cost-Effectiveness $r = +0.48$, Utilization ↔ Cost-Effectiveness $r = -0.03$. The moderate Equity ↔ Cost-Effectiveness correlation (+0.48) is substantively expected (disadvantaged counties also tend to have workforce demand) but does not approach a level that would inflate VIF. The three pillars are statistically independent dimensions, and the 0.40 / 0.35 / 0.25 weights reflect three distinct constructs rather than reshuffled copies of one underlying factor.

Structural-break testing (Chow). A Chow test applied to the 82-month NCDOT BEV registration series identified the August 2022 Inflation Reduction Act passage as a strongly significant structural break ($F =$

1,268.35, $p < 1 \times 10^{-6}$). The May 2025 NCDOT methodology change was also tested, but the result is degenerate ($n_2 = k = 2$: only two post-break observations within the training window, equal to the number of model parameters, producing a mechanically inflated F-statistic that cannot be interpreted). The degeneracy is itself informative: it shows the test window does not yet support inference about that break, and it identifies a future-work re-run when 12 or more months of post-methodology data exist (by mid-2026). The IRA break is the dominant structural feature in the series. The May 2025 methodology change is a minor perturbation (empirical magnitude approximately 0.4% of statewide totals, contributing roughly 20% of the observed underprediction bias) that works *against* the underprediction direction, making the validation a stricter test than the prior methodology would have produced.

Estimator method comparison (CSS versus MLE). SAS Model Studio uses Conditional Sum of Squares (CSS) as its default ARIMA estimator; Python statsmodels uses exact Maximum Likelihood Estimation (MLE) by default. A side-by-side comparison on a statewide ARIMA(1,1,1) produced forecast differences of less than 0.1% across all four-month-ahead horizons (23 to 104 vehicles on a base of approximately 97,000). The AR parameter differed by 0.06%, the MA parameter by 1.48%, and σ^2 by 20% (affecting confidence interval width, not point forecasts). The estimator method choice does not materially affect any result reported in this study.

Confidence interval coverage decomposition. The forecast 95% prediction intervals achieved 75.50% raw coverage on the four-month holdout, below the nominal target. Decomposition explains the gap: observed actuals exceeded the upper bound in 18.0% of observations and fell below the lower bound in only 6.5%, an asymmetric 3:1 ratio indicating a *centering* (bias) problem rather than a *width* problem. After county-specific bias correction (shifting each county's forecast center by its measured bias), CI coverage recovers to 93.75%, essentially at the 95% nominal target. The interval widths are correctly calibrated; the centers are systematically low because BEV adoption accelerated faster than the pre-IRA training data anticipated. The correct policy response is not to widen the intervals but to apply the documented +4 to +5% upward buffer in forecast-based infrastructure allocations (see §4.7's forecast buffer sub-metric).

5. DATA COLLECTION

5.1 NCDOT BEV Registration Data

The demand side uses the complete NCDOT battery electric vehicle registration series: monthly Excel workbooks with county-level BEV, PHEV, hybrid, gasoline, and diesel counts for all 100 North Carolina counties, consolidated by a Python utility (`ncdot_ev_pipeline.py`) into one standardized CSV.

The baseline spans September 2018 through June 2025: 8,200 county-month observations (100 counties, 82 months). A validation extension covering January through October 2025 adds 1,000 records, but January through June 2025 overlaps the baseline, leaving the four-month holdout July through October 2025 as new: 400 records, for a deduplicated total of 8,600. Every county appears in every month, with zero missing values and monotonic growth in statewide BEV totals.

NCDOT changed its counting methodology after May 2025, deduplicating records and excluding recently removed plates (NCDOT ZEV Registration Data page). The structure was unchanged, so the pipeline needed no modification; a flag (`Methodology_PostMay2025`) marks post-change records. The change is roughly 0.4% of statewide totals (about 370 vehicles), with only 10 counties showing a month-over-month decrease at May 2025, none exceeding 5 vehicles. Because it lowers counts, it works against the underprediction seen in validation, a stricter test; Section 4.2 achieved a MAPE of 4.34% against this period.

The BEV column is the primary variable, cumulative rather than new registrations. PHEV registrations are retained for market share but excluded from demand modeling, since NEVI targets public charging and PHEVs can run on gasoline.

5.2 AFDC Infrastructure Data

Supply comes from the U.S. Department of Energy's Alternative Fuels Data Center. The Fall 2025 baseline used a manual CSV filtered to DC fast chargers: 1,725 connector records across 355 stations, 74 columns. It captured nearly all fast-charging ports (1,742 of 1,745 statewide) but missed the 82.4% of stations offering only Level 2 and all 35 Level 1 stations.

A programmatic NREL AFDC API pull replaced it, downloading all operational North Carolina stations (`afdc_api_download.py`). The February 2026 pull captured 1,985 stations with 6,145 total connectors across 267 cities and 358 ZIP codes, comprising 1,876 public and 109 private stations. Level distribution skews to Level 2: 1,636 stations (82.4%) are L2-only, 311 DCFC-only, 29 mixed L2/DCFC, and 9 L1-only. ChargePoint operates the largest share at 51.8% (1,028 stations). Connector types include J1772 (1,586), J1772COMBO (243), Tesla/NACS (234), and CHAdeMO (143), with power output from 2 to 400 kW across the 1,611 stations where power data is available. All subsequent analysis uses the February 2026 pull exclusively.

Cleaning removed 2 non-NC stations (ZIP codes 20020 and 39817), leaving 1,983 NC stations; 56 co-located stations sharing exact coordinates were retained as separate equipment. Join-key validation matched 354 of 358 AFDC ZIP codes to Census ZCTAs (98.9%). The 4 unmatched include the 2 non-NC stations and 2 PO Box ZCTAs (28608 in Boone, 27706 in Durham) with no population record, whose 3 stations (one ZCTA hosting two) were excluded from population-based density calculations. High-missingness fields (EV pricing 86%, facility type 71.5%, owner type 70%) were excluded; core location and port count fields reach 99.8% coverage.

5.3 Census Data

Three Census Bureau products supply population, income, and boundary data. ACS 2022 five-year estimates provide tract-level household income through table B19001 (16 income bins), feeding the workplace charging demand pipeline (Section 4.6) by bridging LODES earnings categories and household income thresholds; ACS also supplies housing tenure (table B25003). Statewide, 44.4% of households report income above \$75,000, and the renter share is 33.8%.

ZCTA population estimates, also ACS 2022, cover 853 North Carolina ZCTAs totaling 10,470,214 residents, 98.8% of the state's approximately 10.6 million. The distribution is right-skewed (mean 12,275 versus median 5,380, skewness 1.70), with 19 uninhabited ZCTAs excluded from per-capita calculations. Population is the denominator in port density (ports per 10,000 residents) and the population-weighted Gini weight.

County and ZCTA boundary shapefiles come from Census TIGER/GENZ 2020 at 500k resolution. All 100 county geometries are valid and use NAD83 (EPSG:4269); the 853 NC-prefix ZCTAs were filtered via spatial join. Census tract boundaries in the 2010 vintage were required separately for the CEJST equity crosswalk, since CEJST v2.0 used 2010 tract definitions and no successor migration has occurred (see Section 5.5).

5.4 LEHD LODES Commuter Data

Workplace charging demand uses the Census Bureau's LEHD Origin-Destination Employment Statistics (LODES) dataset, version 2021, providing block-level origin-destination commuter flows. The North

Carolina origin-destination main file contains 3,768,428 intra-state flow records with zero nulls and zero duplicate block pairs; a Workplace Area Characteristics (WAC) file contains 71,921 workplace blocks.

LODES segments workers into SE01 (below \$1,250/month), SE02 (\$1,251 to \$3,333/month), and SE03 (above \$3,333/month, roughly \$40,000 annually). SE03 is the initial filter for EV-affordable commuters, reducing the pool from 4,198,163 total workers to 2,041,731. Additive decompositions (SA01+SA02+SA03=S000, SE01+SE02+SE03=S000, SI01+SI02+SI03=S000) verified for all origin-destination records, and CE01+CE02+CE03=C000 for all WAC records; a block-to-county crosswalk achieved 100% coverage, all 100 counties present.

Alternatives were rejected: the Census Transportation Planning Products (CTPP) dataset for older vintage (2012-2016 versus LODES 2021), county-level geography, and restricted access; StreetLight Data and Replica for \$10,000 to \$100,000 costs; and the National Household Travel Survey for state-level-only estimates.

5.5 CEJST / Justice40 Data

The Climate and Economic Justice Screening Tool version 2.0 (CEJST v2.0) was developed by the Council on Environmental Quality (CEQ) and was accessible at screeningtool.geoplatform.gov. It was the federal instrument for identifying disadvantaged communities under the Justice40 Initiative (Executive Order 14008) from its December 2024 release through January 22, 2025. Its methodology classifies a Census tract as 'disadvantaged' if it is both low-income (at or above the 65th percentile for households below 200% of the federal poverty level) and overburdened on at least one of eight environmental, health, or infrastructure dimensions. This is documented in the CEJST v2.0 Technical Support Document [6], which remains the federal record of the criteria independent of the tool's operational status.

The dataset covers 2,195 North Carolina Census tracts at 2010 boundaries, of which 934 of 2,170 evaluable tracts (43.0%) carry the disadvantaged designation after excluding 25 zero-population tracts (12 water-body, 13 institutional and military). The 43.0% rate exceeds the national average of approximately 37%.

The tool was removed from screeningtool.geoplatform.gov on January 22, 2025 following the rescission of Executive Order 14008; the EDGI / Public Environmental Data Partners (PEDP) coalition archived and restored the data within days. This study uses the community-preserved CEJST v2.0 (December 2024 release) as a bulk CSV from the PEDP mirror, verified against pre-removal records, with the CEQ Technical Support Document preserved independently. The archival pathway is therefore a *data continuity* matter, not a methodological one: the methodology is fully reproducible from the federal record, and the data preserved by an independent coalition with chain-of-custody documentation. No API exists for CEJST; bulk CSV is the only access path.

No federal successor exists at this writing; the EPA EJScreen tool uses different burden categories and threshold percentiles and is not a drop-in replacement. CEJST-anchored equity findings should be re-examined against any successor. Section 3.4 discusses the broader policy context.

5.6 Data Quality and Limitations

Temporal mismatch is the most consequential limitation: NCDOT runs through October 2025, AFDC reflects February 2026 status, ACS uses 2022 five-year averages (2018–2022 responses), LODES reports 2021 commuting, and CEJST draws on federal datasets with varying reference periods. No two sources share the same window.

The LODS vintage carries a specific concern: 2021 captured commuting when COVID-era remote work was still elevated. The 0.85 remote work multiplier applied in the demand pipeline (Section 4.6) adjusts for this, but it is a national average from Barrero, Bloom, and Davis [1], not a North Carolina-specific estimate; the project brackets the uncertainty with low, baseline, and high scenarios.

CEJST's 2010 tract boundaries against 2020-vintage ZCTAs and counties require a crosswalk. At the county level the mismatch is negligible, since North Carolina's boundaries are identical between vintages. At the sub-county level, the area-weighted interpolation in the tract-to-ZCTA crosswalk (Section 4.5) assumes uniform within-tract population, introducing modest error in large rural tracts.

One gap shaped scope from the outset: NCDOT does not publish ZIP-code-level BEV counts. The ZIP-code density analysis (Section 4.4) measures supply-side inequality using port density per 10,000 residents but cannot compute BEV-to-port ratios at the ZIP level, since the demand denominator does not exist there. Commercial alternatives (IHS Markit, Experian VIO) require licenses of \$5,000 to \$20,000 per state. The analysis uses county-level BEV counts for demand and ZIP-level port counts for supply.

These limitations are taken up systematically in Section 9 (Limitations), alongside three federal-policy-specific limitations (CEJST methodology lock at v2.0, NEVI program rule evolution risk, and state-administered formula program variance) and the direction-of-bias analysis accompanying each.

6. RESULTS

6.1 Forecast Validation

Out-of-sample validation tested SAS Model Studio's county-level BEV forecasts against four months of unseen data (July through October 2025). The mean absolute percentage error across all 100 counties was **4.34%**, below the 5% usability threshold. Three model families contributed: exponential smoothing (82 counties), ARIMA (13), and unobserved component models (5). Figure 44 plots actual versus predicted registrations across all 400 county-month observations; points above the 45-degree line show the systematic underprediction discussed below.

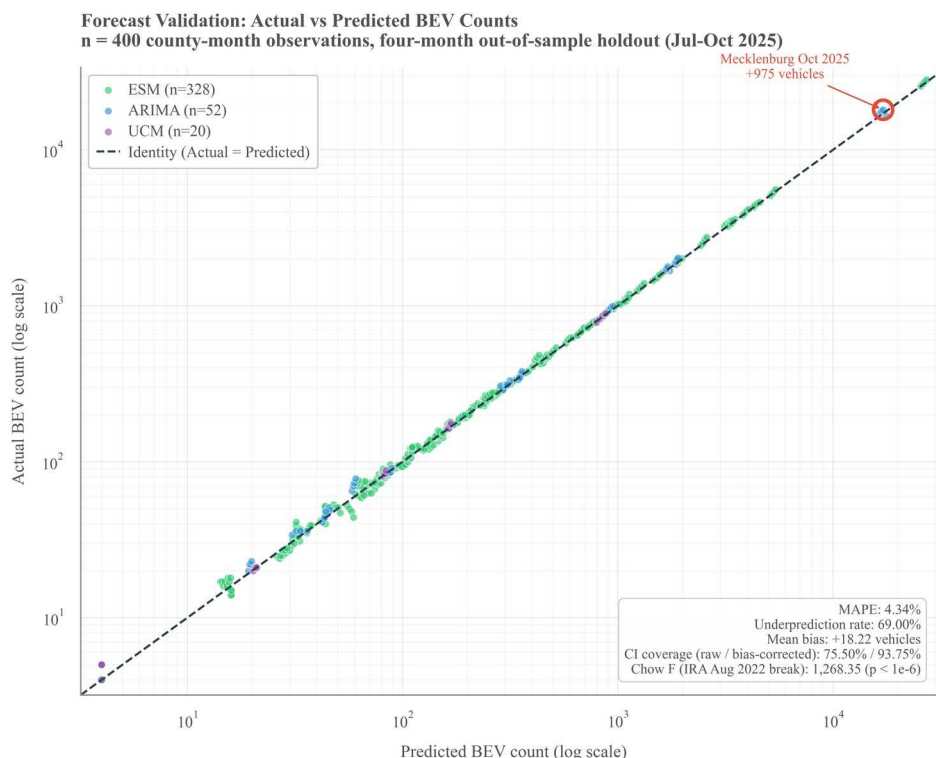


Figure 44. Forecast Validation: Actual Versus Predicted BEV Registrations Across 400 County-Month Observations

Actual registrations exceeded forecasts in **69.00%** of observations, with a mean bias of **+18.22** vehicles per county-month. The 95% prediction intervals achieved **75.50%** raw coverage. As decomposed in §4.9, this reflects a *centering* (bias) problem rather than a *width* problem: after county-specific bias correction, coverage recovers to **93.75%**. A Chow structural-break test on the 82-month series confirms a real adoption acceleration: $F = 1,268.35$ ($p < 1 \times 10^{-6}$) at the August 2022 Inflation Reduction Act passage. The models learned historical patterns now too conservative; Mecklenburg's October 2025 actual exceeded its forecast by 975 vehicles in a single month.

The underprediction has a concrete policy consequence: forecasts used without adjustment would undersize deployment by roughly 4 to 5%. The study carries this buffer into the scoring equation's utilization pillar, applied globally because county-specific estimates over four months would overfit.

Table 1

Headline Findings

#	Finding	Value	Section
1	Out-of-sample forecast validation (MAPE, n = 400 county-months)	4.34%	§6.1
2	Forecast underprediction rate	69.00%	§6.1

3	IRA structural break (Chow F-statistic, $p < 1 \times 10^{-6}$)	1,268.35	§4.9
4	CI coverage (raw / county-specific bias- corrected)	75.50% / 93.75%	§4.9
5	Statewide BEV-to-port ratio (NC, Feb 2026; IEA global benchmark ≈ 10)	16.9	§1.2, §6.4
6	County-level BEV ownership Gini	0.805	§1.2, §6.3
7	ZIP-level infrastructure Gini (statewide, population-weighted)	0.566	§6.3
8	Theil-T within-county share of inequality (top- 10 cohort)	84.5%	§6.3
9	Justice40-designated NC Census tracts (CEJST v2.0)	43.0% (934 of 2,170)	§3.4, §6.5
10	Stations sited in Justice40 tracts (top-10 county scope, as of Jan 21, 2025)	24.5% (296 of 1,210)	§6.5
11	Top-3 NEVI counties (stable across weight sensitivity)	Union (0.561), Mecklenburg (0.548), Guilford (0.465)	§6.6
12	Scoring pillar independence (maximum VIF; concern threshold 5.0)	1.41	§4.9

Note. The table consolidates the most-cited numerical findings of this study for readers who may engage at a skim level. Section pointers point to where each finding is established.

6.2 Infrastructure Coverage Gaps

The February 2026 AFDC API pull shows an infrastructure picture substantially different from the baseline's DCFC-only extract. Of 853 Census ZCTAs in North Carolina, 499 (58.5%) have zero charging stations of any type, representing 2.2 million people, or 21.1% of the state's population. The gap is not

rural alone; 9 of the 10 largest counties by BEV count have at least one ZIP in the top 20 most underserved list.

Within the top 10 counties, 134 ZIP codes serve 4.4 million residents with 3,722 charging ports. Mean density is 11.18 ports per 10,000 population, but the median drops to 5.51. Charlotte Uptown (28202) sits at 78.64 ports per 10,000 people; fourteen miles east, Charlotte ZIP 28215 has 64,713 residents served by 2 ports, a density of 0.31, a 250-fold gap within the same county. Burlington (27215) fares worse absolutely: 1 port for 45,982 people, a density of 0.22 per 10,000, roughly 50 times below the statewide median.

The top 20 underserved ZIPs hold 732,892 residents served by 58 ports, and 15 have zero DC fast chargers. Mecklenburg and Guilford each contribute 5 ZIPs to the top 20, so the state's two most infrastructure-rich counties also hold some of its worst gaps. Several underserved ZIPs have chargers that are private access only (Winston-Salem 27105), making their effective public supply zero.

6.3 Intra-County Inequality

The Gini coefficient on ports per 10,000 population across populated ZCTAs gives a statewide population-weighted value of 0.566, qualifying as very high inequality. Mecklenburg tops the list at 0.623, followed by Guilford at 0.571; Durham records the lowest among the top 10 at 0.408. Durham's weighted Gini (0.408) diverges from its unweighted value (0.51) because the well-served ZIPs are the populated ones; in Mecklenburg no such divergence appears, as large-population ZIPs are underserved alongside small ones.

Union County is easy to misread: its low Gini might suggest equity, but every ZIP is equally underserved, with an interquartile range of port density of just 0.92. Equality at a low level of service is not adequate access.

The Gini cannot decompose inequality spatially. The Theil-T (GE(1)) index, exactly decomposable into between- and within-group components via the additive decomposability of the GE(1) class [2, 14], answers what the Gini leaves open. Total Theil-T for the top 10 counties is 0.5791: 0.0900 (15.5%) falls between counties and 0.4892 (84.5%) within counties. A Theil-L (GE(0)) check confirmed the pattern at 82.5% within-county, both decompositions verifying to machine precision (difference of 2.22×10^{-16} between summed components and total). Figure 33 presents this as a stacked bar chart.

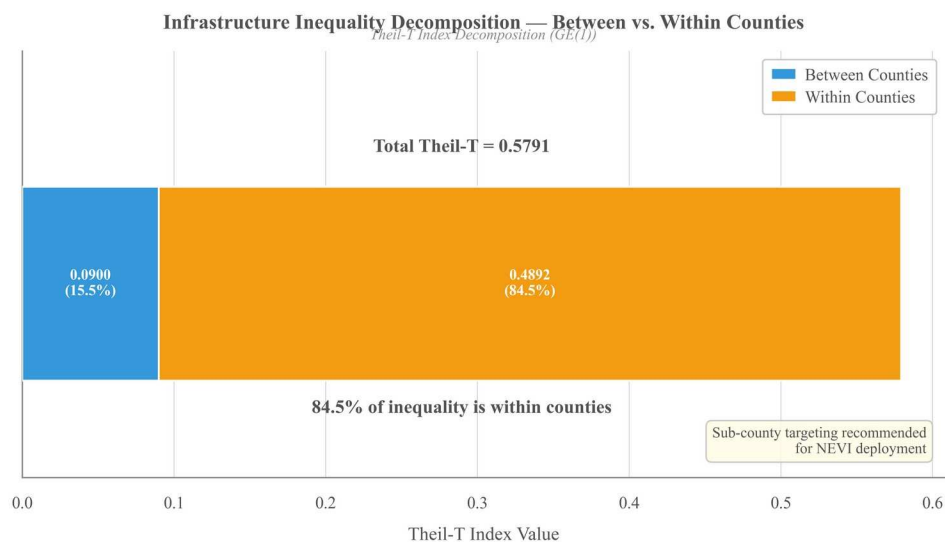


Figure 33. Theil-T Decomposition of EV Charging Infrastructure Inequality: Within-County vs Between-County

Mecklenburg alone contributes 41.5% of within-county inequality and Wake 28.1%, together roughly 70%, driven by large populations and extreme internal variation. Guilford, despite the second-highest Gini, contributes only 7.7% because its population is smaller, and Union 0.6%, consistent with its uniform-but-low service.

The 84.5% within-county finding has a direct policy implication: county-level allocation formulas, the basis of most state NEVI programs, would miss the majority of the inequality problem. Figure 24, the Mecklenburg heat map, makes this visible: Charlotte Uptown is a dark cluster of concentrated infrastructure surrounded by pale green and hatched zero-station areas including 28215 and its 64,713 residents. That is what a Gini of 0.623 looks like on a map.

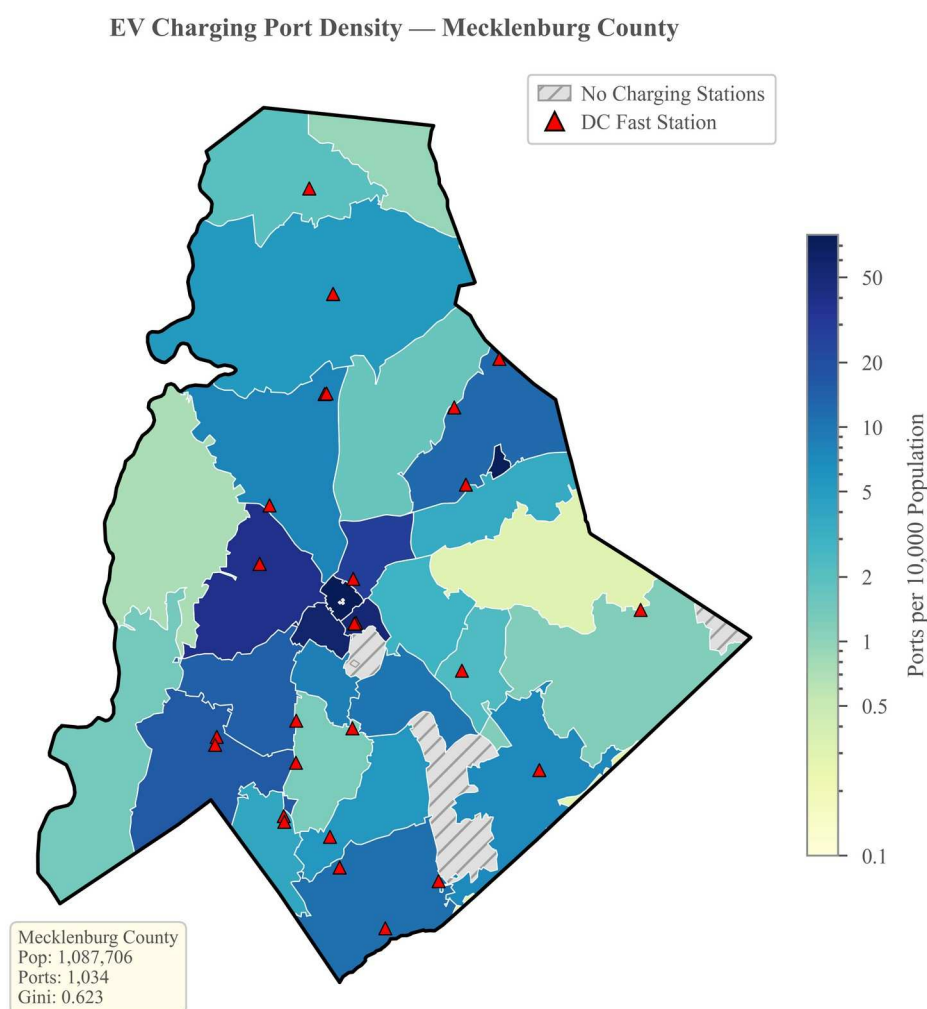


Figure 24. Mecklenburg County: Charging-Station Density Heat Map

6.4 Workplace Charging Demand

Layering commuter flows onto residential registration data changes the demand picture substantially. Mecklenburg absorbs +194,361 net commuters per day, Wake +126,517, and Durham +89,450. After the

three-layer adjustment pipeline (SE03 income filter, ACS income correction to the \$75,000 household threshold, and 0.85 remote work reduction), statewide adjusted workplace demand totals 859,260 workers. The pipeline filters from 4,198,163 raw workers to 2,041,731 after SE03 selection (48.6%), to 1,010,894 after the income correction (49.5% of SE03), to 859,260 after the remote work multiplier.

Net commuter flows classify all 100 NC counties into 8 employment centers (net inflow above +10,000), 75 balanced counties, and 17 bedroom communities (net outflow below -10,000). Union, first in the NEVI scoring, is a bedroom community of Charlotte with a net outflow of -36,113 commuters per day; its residents drive to Mecklenburg for work, so Union's residential BEV-to-port ratio understates its workday need and Mecklenburg's understates its daytime demand from non-residents.

At baseline assumptions (30% charging adoption rate, 15:1 port ratio), statewide workplace port need is 17,185 ports (13,748 Level 2, 3,437 DCFC), with Mecklenburg needing 3,658, Wake 3,391, and Durham 1,496. For context, NC's statewide public BEV-to-port ratio is 16.9 (NCDOT registrations divided by AFDC ports as of February 2026), against the International Energy Agency's global benchmark of approximately 10 BEVs per port [8]. Workplace charging efficiency feeds the cost-effectiveness pillar, where Mecklenburg scores highest (0.801) and Buncombe lowest (0.034), reflecting a 194,000-commuter employment center versus a tourism-driven economy. Figure 36 compares residential-only and residential-plus-workplace demand for the 10 scoring counties.

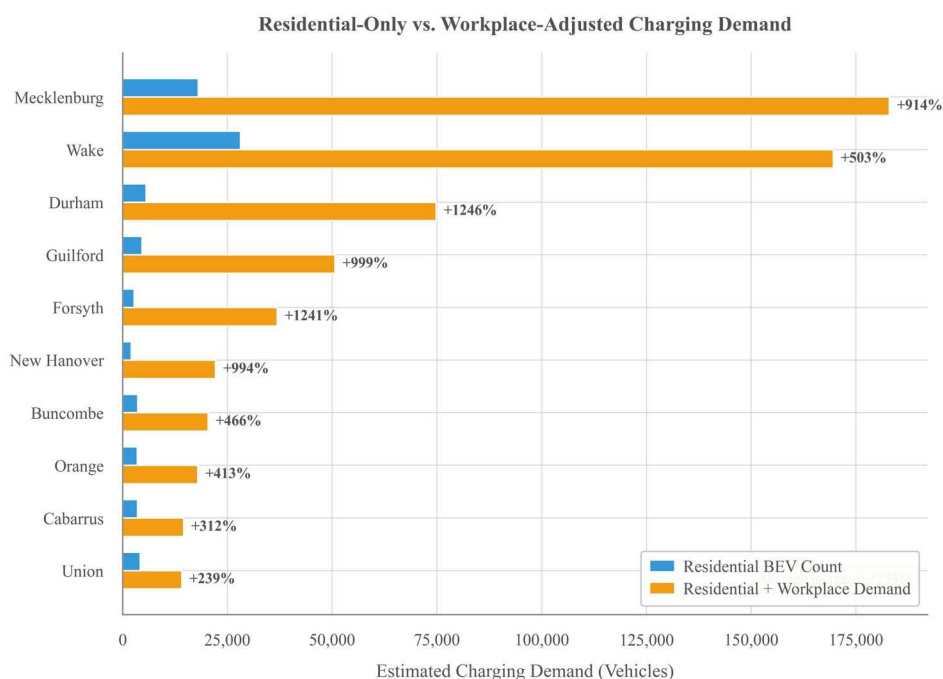


Figure 36. Workplace Charging Demand: LODES-Adjusted vs Unadjusted Estimates

6.5 Equity Analysis

CEJST v2.0 designates 934 of 2,170 evaluable North Carolina Census tracts (43.0%) as disadvantaged under the Justice40 definition, above the national average of approximately 37%. The elevated rate reflects rural poverty in eastern NC and Appalachia, above-average health burdens, housing cost stress, and legacy industrial contamination. Within the top 10 BEV counties the population-weighted disadvantaged rate drops to 18.5%, expected for North Carolina's wealthiest urban and suburban counties. County rates vary widely: Guilford 29.2%, New Hanover 28.8%, Durham 26.6%, Mecklenburg 23.7% (the largest absolute count), and Wake only 8.1% despite the highest BEV registrations.

A sensitivity analysis removed CEJST's most debatable component, climate change. Of 934 disadvantaged tracts, 124 (13.3%) are flagged solely by climate indicators; removing them drops the statewide rate from 43.0% to 37.3%. The impact on study counties is minimal: 7 of 10 are unaffected, and the three affected (New Hanover -4.7 percentage points, Durham -3.4, Wake -2.2) shift modestly. The equity rankings are driven by empirical health, housing, and workforce indicators, not climate projections.

A spatial overlay of station locations on CEJST-designated tracts quantifies how infrastructure aligns with Justice40 communities. Of the 1,210 stations within the top-10 study county scope, **296 (24.5%) sit in census tracts designated disadvantaged under CEJST v2.0 as of January 21, 2025**, the last operational day of the federal screening tool before its removal. The 24.5% station share sits 18.5 percentage points below the 43.0% statewide share, but the more comparable benchmark is the 18.5% population-weighted disadvantaged rate within the study counties. Against that benchmark, the station share runs approximately proportional with a modest surplus (24.5% vs 18.5%) but strong county-by-county variation. Some counties site stations roughly in proportion to their disadvantaged-tract population; others show pronounced under-representation that drives the equity component of their NEVI scores.

Figure 42, overlaying station locations on Justice40-designated tracts, is the most legible output of Phase 5: the disconnect between where stations cluster and where disadvantaged populations live requires no statistical interpretation.

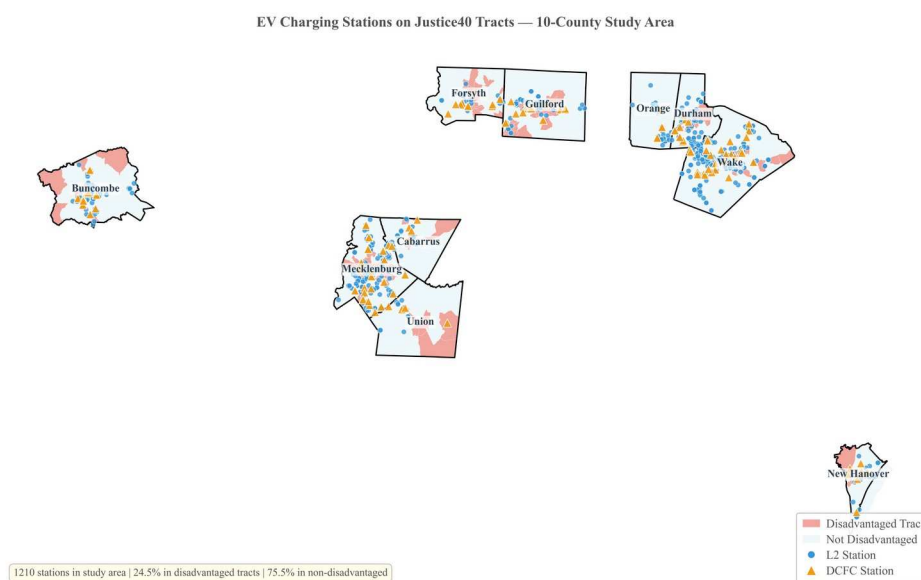


Figure 42. Charging Stations Overlaid on Justice40-Designated Disadvantaged Tracts

6.6 NEVI Priority Scores

The scoring equation described in Section 2.3 produces a single composite ranking for each of the 10 study counties. The full results, computed from `scoring_framework_final.py` and verified against the NEVI equation arithmetic to within 1×10^{-12} (Section 4.9), are shown in Table 2.

Table 2

NEVI Priority Scores for the Top 10 NC Counties by BEV Registration

Rank	County	NEVI Score	Equity	Utilization	Cost-Eff.
------	--------	------------	--------	-------------	-----------

1	Union	0.561	0.319	1.000	0.333
2	Mecklenburg	0.548	0.810	0.067	0.801
3	Guilford	0.465	0.855	0.103	0.347
4	New Hanover	0.341	0.607	0.071	0.293
5	Wake	0.316	0.322	0.130	0.568
6	Durham	0.313	0.469	0.033	0.454
7	Forsyth	0.285	0.283	0.156	0.467
8	Cabarrus	0.199	0.228	0.229	0.111
9	Buncombe	0.197	0.470	0.000	0.034
10	Orange	0.077	0.059	0.101	0.071

The top three reach high scores through different pathways. Union is utilization-driven: its 101.5 BEV-per-port ratio is more than three times the next-highest county (Cabarrus at 32.0), producing a utilization score of 1.000 in a suburban corridor where deployment has not kept pace with adoption. Mecklenburg is equity-driven: its equity sub-score of 0.810 is second-highest (behind Guilford at 0.855), and its cost-effectiveness of 0.801 reflects the scale of its employment center. Guilford combines the highest equity burden with moderate utilization strain, placing it third.

Wake ranks fifth despite the most BEV registrations in the state. Its equity score of 0.322 reflects a low Justice40 burden (only 8.1% of tracts disadvantaged), and while its cost-effectiveness is strong (0.568, second overall), the 0.40 equity weight pulls the composite down. The equation weights equity consistent with the Justice40-aligned federal benchmark rather than rewarding raw demand.

Orange sits last at 0.077, low across all three pillars: a Justice40 population-weighted burden of 4.9%, adequate per-capita infrastructure, and a university-town demographic with a small, well-served EV fleet.

Weight sensitivity analysis tested equity weights from 0.30 to 0.50 in 0.05 increments while holding the utilization-to-cost-effectiveness ratio at 7:5. The top-three set (Union, Mecklenburg, Guilford) held across all five scenarios; internal ordering shifted predictably, with Mecklenburg gaining and Union falling as equity weight rises, and Orange steady at tenth. The rankings are not artifacts of a particular weighting choice (see Section 2.3 for the weight rationale; combined with the cost-effectiveness sub-weight robustness and the remote-work multiplier cancellation in min-max normalization, this is the "triply-robust" top-3 finding referenced in §4.7.1).

6.7 NEVI Priority Score Visualization

Figure 43 visualizes the Table 2 rankings as a horizontal stacked bar chart, each county's total NEVI Priority Score decomposed into its three pillar contributions (Equity at 0.40, Utilization at 0.35, Cost-Effectiveness at 0.25). The archetypes are immediately legible. Union's bar is dominated by Utilization (the 1.000 normalized score driving the #1 ranking despite moderate equity), Mecklenburg's shows

roughly balanced Equity and Cost-Effectiveness, Guilford's is dominated by Equity (the cohort's highest sub-score at 0.855), and Orange's short bar reflects low scores across all three pillars. The figure uses the Okabe-Ito colorblind-safe palette, sorts counties by total score descending, annotates each bar to three decimals matching Table 2, and renders at 600 DPI.

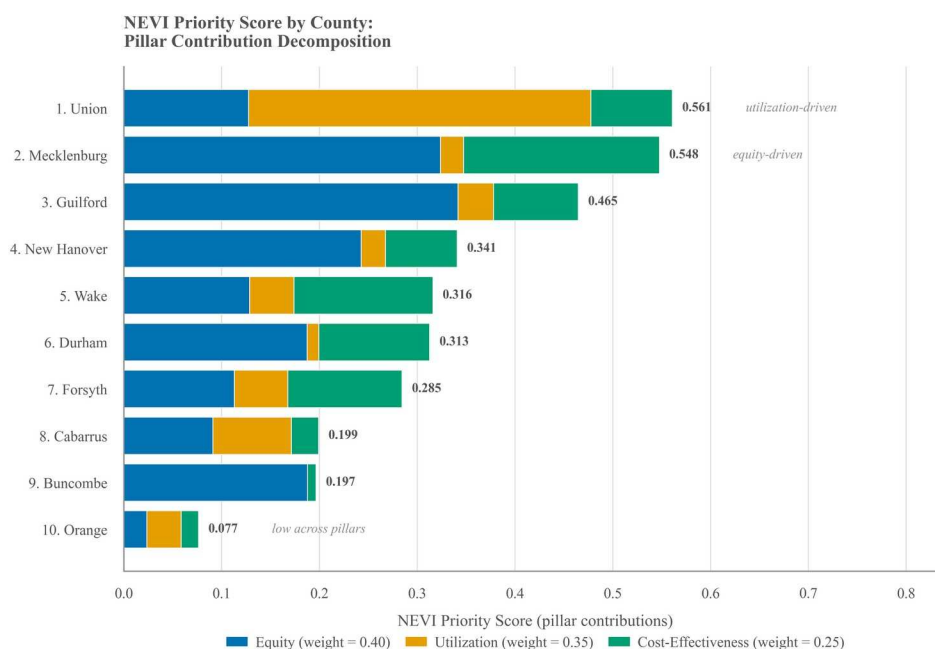


Figure 43. NEVI Priority Scores: Top-10 NC Counties

7. DISCUSSION

This section synthesizes the findings through three frames: a cross-lane methodology synthesis substantiating the NEVI Priority Score's defensibility; a stakeholder mapping of what the framework delivers; and a three-archetype interpretation that converts the top-10 ranking into a policy conversation.

7.1 Cross-Lane Methodology Synthesis

Four independent lanes converge on the rankings in Section 6.6 (Figure 43). In data engineering, the CEJST tract-to-ZCTA crosswalk validation suite (12 checks in 3 tiers, 23 of 23 sub-checks passing) is the pipeline's best-validated component. The 0.40 equity weight therefore rests on geometry tested to federal precision (Section 4.5), backed by a consolidated data dictionary, a 19-entry provenance table, a reproducibility lockfile (`uv.lock`), and 27 analysis scripts. Spatially, every area-dependent operation uses NC State Plane (EPSG:32119, meters) and every topological-only operation WGS84 (EPSG:4326), so no area or length calculation occurs in a geographic CRS. The 2010-to-2020 tract-vintage mismatch is immaterial at the county level because NC county boundaries are identical between the two Census decades, with a per-ZCTA area-conservation check (within 1%) backstopping the claim. On federal compliance, the Data Governance Readiness chain (Section 5.6) and consolidated provenance (Section 5) establish end-to-end traceability through immutable raw data, acquisition scripts, the validation suites (MAPE 4.34%, crosswalk 23/23, Phase 4 EDA 59/59), scored outputs, and the lockfile. The CEJST archival pathway is disclosed (Section 5.5), the federal screening tool decommissioned January 22, 2025 while the community archive and CEQ Technical Support Document preserve data and methodology. Three documented limitations each favor the analysis: the NCDOT methodology change works against the underprediction (Section 5.1), interstate commuter exclusion undercounts workplace demand, and the

remote-work multiplier cancels in normalization (Section 4.7.1). Econometrically, the MAPE of 4.34% on a true four-month out-of-sample holdout (Section 6.1) supports county-level capital allocation. The maximum Variance Inflation Factor of 1.41 across the three pillars (Section 4.9) means the 0.40 / 0.35 / 0.25 weights reflect three statistically independent dimensions of need. The Chow F-statistic of 1,268.35 at the August 2022 IRA passage date (Section 4.9) converts the 69.00% underprediction into the fingerprint of an accelerating adoption regime, making the gap a floor, not a ceiling. The triply-robust top-3 finding (Section 4.7.1) holds Union, Mecklenburg, and Guilford across the equity-weight sweep, the cost-effectiveness sub-weight robustness, and the remote-work cancellation.

7.2 Stakeholder Implications

NCDOT and FHWA (NEVI administrators). A ranked list of 10 counties (Union, Mecklenburg, Guilford, New Hanover, Wake, Durham, Forsyth, Cabarrus, Buncombe, Orange) whose top-3 ordering survived every robustness test, whose diagnostics are defensible (VIF 1.41, 4.34% MAPE), and whose equity geometry passes a 23-of-23 validation suite. The list covers 73% of NC's BEV fleet; statewide extension is future work (Section 11). The deliverable is a priority ranking, not a dollar allocation.

Federal Justice40 auditors and OIG-style reviewers. The crosswalk aligns to EPA EJScreen and HUD USPS Crosswalk standards (Section 4.5). The CEJST v2.0 archival pathway is disclosed with verified schema match and a CEQ anchor (Section 5.5), and a climate sensitivity analysis leaves 7 of 10 study counties unchanged even if the most-contested CEJST category is removed (Section 4.5). The Theil-T decomposition (84.5% within-county inequality; Section 6.3) reframes equity as intra-urban. The 0.40 equity weight is statistically independent of the other pillars (Section 4.9) and reflects the Justice40 target under Executive Order 14008, treated as an analytical anchor given the post-rescission landscape (Section 3.4).

County and regional planners. The sub-county (ZIP-level) gap analysis is the most actionable output. The 250-fold density gap between Charlotte ZIP 28202 (78.64 ports per 10,000 residents) and Charlotte ZIP 28215 (0.31 ports per 10,000), 14 miles apart in Mecklenburg, is the operational form of the Theil decomposition (Section 6.3). The three archetypes (Section 7.3) tell planners which intervention fits their county type.

Private-sector charging-network operators. A competitive baseline anchored on the February 2026 AFDC API pull (1,985 stations, 6,145 connectors), a market-gap number (NC statewide 16.9 BEVs per public port versus the IEA Global EV Outlook 2024 benchmark of approximately 10), and workplace-charging sizing from LEHD LODES flows (859,260 adjusted commuter demand statewide).

Academic and research community. A reproducible methodology backed by a Claims-to-Evidence Matrix (Appendix A) and a dependency lockfile. The contribution is the first additive Theil-T decomposition of EV charging infrastructure inequality applied to a state-level allocation problem (Section 3.3), supported by the tract-to-ZCTA crosswalk and CEJST climate sensitivity analysis (Section 4.5); all five Unique Contributions in Section 10 are substantiated.

Public and Justice40 communities. Procedural and distributional fairness: a transparent, published scoring formula with the largest weight on equity (0.40), a disclosed scope (73% of NC BEV fleet, top-10 counties), and fully public, reproducible data traceable via the Claims-to-Evidence Matrix (Appendix A).

7.3 Three County Archetypes

The top-10 ranking in Section 6.6 sorts by composite NEVI Priority Score but does not say *why* each county sits where it does. Figure 45 visualizes the framing as a scatter in (Equity × Utilization) pillar space, bubble size proportional to the composite score and quadrant shading naming each archetype.

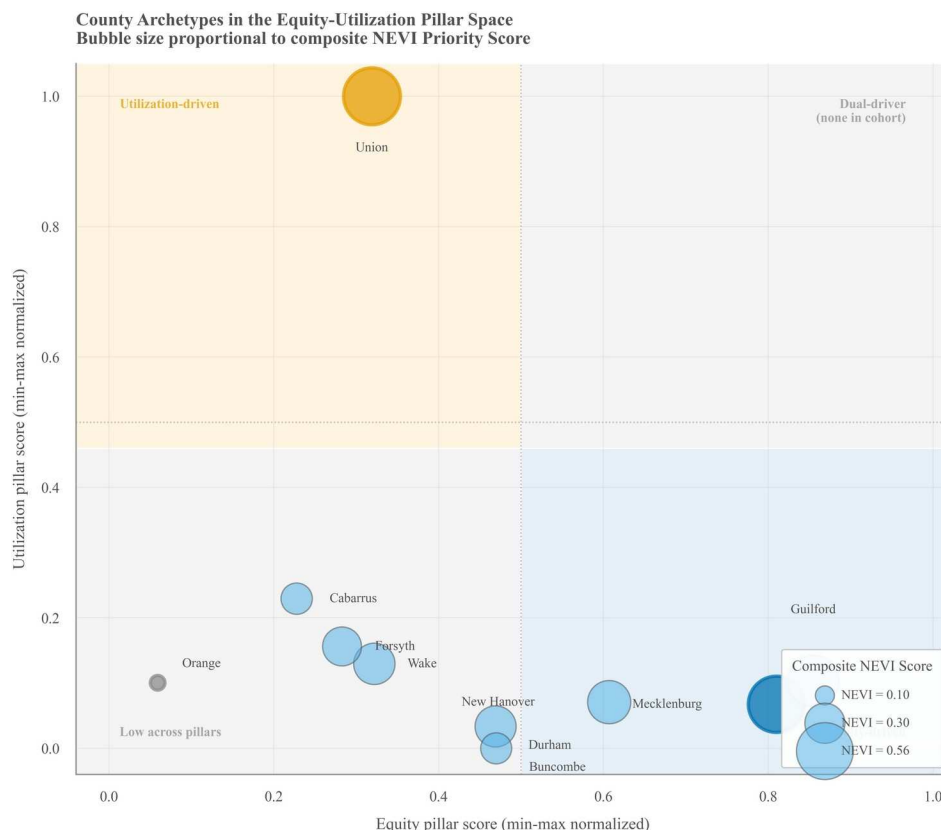


Figure 45. Equity-Utilization Quadrant: Three County Archetypes

Union County (Rank 1, NEVI = 0.561): utilization-driven. Union's 101.5 BEV-per-port ratio is over three times the next-highest county (Cabarrus at 32.0), yielding the maximum normalized utilization score of 1.000 (Section 6.6). A Charlotte bedroom community with a net commuter outflow of -36,113 per day, its #1 ranking reflects residential under-deployment: the implication is *more stations*.

Mecklenburg County (Rank 2, NEVI = 0.548): equity-driven. Mecklenburg's equity sub-score of 0.810 is second-highest (behind Guilford at 0.855), paired with the highest cost-effectiveness sub-score (0.801), driven by 194,361 net commuters per day at Charlotte's employment centers. Its ranking pairs with the 250-fold within-county density gap (Charlotte Uptown 78.64 versus Charlotte East 0.31; Section 6.2) and contributes 41.5% of within-county inequality in the cohort (Section 6.3). The implication is *better-targeted stations* in underserved disadvantaged ZIPs. Guilford County (Rank 3, NEVI = 0.465) sits in the same archetype with the cohort's highest equity sub-score and moderate utilization strain.

Orange County (Rank 10, NEVI = 0.077): low across all pillars. Orange's Justice40 disadvantaged-tract burden is 4.9%, its per-capita infrastructure is adequate, and its university-town demographic produces a small, well-served EV fleet. Orange ranks last across all five tested equity weight scenarios (0.30 to 0.50), confirming the position is structurally stable.

The fourth quadrant in Figure 45 (high equity *and* high utilization) is empty: no NC county combines high disadvantaged-tract burden with extreme under-deployment. The empty quadrant is itself a finding: every cohort county presents a policy trade-off the framework makes explicit rather than dissolving.

7.4 Geographic Sensitivity: Coastal versus Inland

One additional pattern emerged from the climate sensitivity analysis (Section 4.5). New Hanover County (Rank 4) showed the highest sensitivity to removal of the CEJST climate change category, its disadvantaged-tract rate dropping 4.7 percentage points. Durham (-3.4 percentage points) and Wake (-2.2 percentage points) showed smaller but non-zero sensitivity, while the remaining seven study counties showed zero impact. The pattern tracks New Hanover's coastal exposure via FEMA NRI hazard projections. The implication is narrow: coastal NC siting decisions are more sensitive than inland ones to the choice of equity screening methodology, but the effect is bounded and does not shift the top-3.

7.5 Synthesis

The methodology lanes (§7.1) substantiate the Section 6.6 rankings statistically, geometrically, and via traceability; the stakeholder mapping (§7.2) gives each constituency a defensible artifact for its decision context. The archetype framing (§7.3, Figure 45) converts the ordered ranking into a policy interpretation county-level aggregation would obscure, and the geographic sensitivity analysis (§7.4) bounds the coastal-versus-inland framing without disturbing the top-3. Together they supply the prescriptive layer policymakers can use to inform deployment decisions, not to substitute for them.

8. PROJECT EVALUATION

8.1 Project Outcomes

The project produced six artifact classes. First, a county-level NEVI Priority Score for the top 10 North Carolina counties by BEV registration count, with rankings stable across the equity-weight sensitivity sweep, the cost-effectiveness sub-weight robustness test, and the remote-work multiplier cancellation (Section 4.7.1). Second, a sub-county ZIP-level infrastructure gap analysis usable for site selection, with the Theil-T decomposition (Section 6.3) establishing the within-county dominance that justifies the two-tier architecture. Third, a reproducible analytical pipeline of 27 documented Python scripts, a dependency lockfile, and three validation suites (MAPE 4.34% out-of-sample, crosswalk 23 of 23, Phase 4 EDA 59 of 59) re-executable from a clean clone. Fourth, a Claims-to-Evidence Matrix tying every numerical assertion to a source file, packaged with the Data Quality Report (Appendix A). Fifth, two documented internal methodology reviews. Sixth, a publicly released analysis repository under an open noncommercial license with 42 publication-grade figures. Each artifact is positioned for use rather than display: the rankings inform allocation rather than dictate it, and the components are independently auditable.

8.2 Contribution to Stakeholders

The work serves six stakeholder groups: NEVI program administrators (NCDOT and FHWA), Justice40 compliance reviewers, local and regional planners, private-sector charging operators, the academic community, and the Justice40 communities the program serves. To administrators, it supplies a county-level priority ranking whose statistical defensibility (VIF 1.41, MAPE 4.34%, triply-robust top-3) is appropriate to the funding decision it informs, as a prescriptive layer rather than a replacement for program deliberation. To auditors, a 23-of-23 federal-standard crosswalk validation with traceable CEJST v2.0 data lineage (Section 4.5). To planners, sub-county granularity (Section 6.2) that makes the Theil-T finding operationally usable. To the research community, a reproducible pipeline and the first additive Theil-T decomposition applied to EV charging infrastructure inequality (Section 10.3). To Justice40 communities, a public, reproducible methodology in which the equity pillar carries the largest weight (0.40). The methodological learning surfaced and the sole-author execution self-assessment are documented in Appendix A.

9. LIMITATIONS

Each limitation names a direction of bias (against the headline finding, toward it, cancels mathematically, or bounded) and a future-work pointer, grouped as reinforcing the findings (§§9.1–9.4), bounded or undetermined (§§9.5–9.7), and federal-policy or program risks (§§9.8–9.10).

9.1 Top-10 County Selection Limits Generalizability

The framework covers the top 10 NC counties by BEV registration count, not all 100 (73% of the statewide BEV fleet), over-representing urban/suburban dynamics. The Theil-T decomposition (Section 6.3) shows 84.5% of charging infrastructure inequality lives within counties, which the scoring targets. *Direction of bias: bounded by the Theil finding and the weight sensitivity analysis. Future work: extension to all 100 counties, activating the dormant sub-metrics (Section 9.5).*

9.2 LODS 2021 Vintage Predates Full Return-to-Office

The workplace demand model (Section 4.6) uses LEHD LODS 2021 commuter flows from a high-remote-work period; commercial alternatives cost \$10,000 to \$100,000 per state. The 0.85 remote-work multiplier [1] applies the national average. *Direction of bias: cancels mathematically in min-max normalization.* Rankings are identical at 0.75, 0.85, and 0.95 (Section 4.7.1); the vintage shifts absolute demand (e.g., the 859,260 statewide figure in Section 6.4) but not ordering. *Future work: county-specific rates from post-COVID ACS.*

9.3 CEJST One-Strike Design May Over-Identify Disadvantaged Tracts

CEJST v2.0 classifies a tract as disadvantaged if it exceeds the 90th percentile on any single indicator across eight burden categories, given the low-income threshold (Section 3.4). Removing the climate change category drops the 43.0% NC rate to 37.3% (Section 4.5); seven of ten counties show zero impact, and the three affected (New Hanover, Durham, Wake) shift by 2.2 to 4.7 percentage points. *Direction of bias: bounded; reinforces robustness, as rankings rest on health, housing, and workforce indicators. Future work: re-examine against any successor instrument.*

9.4 Static BEV Registration; No Growth Trajectory Modeling

The framework treats BEV registrations at October 2025 levels, not modeling continued growth. Phase 1 validation (Section 6.1) reports MAPE 4.34% on a four-month holdout, the 69.00% underprediction rate explained by the August 2022 IRA structural break (Chow F = 1,268.35; Section 4.9). *Direction of bias: under-states demand growth, making BEV-per-port scores conservative. Future work: longitudinal modeling on post-IRA windows; ESM/ARIMA/UCM ensembles.*

9.5 Two Zero-Variance Sub-Metrics at 10-County Scope

Two of nine sub-metrics carry zero variance here: `zero_station_pct` is 0.0 for all 10 counties, and `forecast_buffer` is a uniform 0.045 applied globally (Section 4.7). *Direction of bias: no directional effect; bounded scope. Future work: statewide extension activates both.*

9.6 No Charging Behavior Data Beyond AFDC

The Alternative Fuels Data Center provides station-level location and access type, not real-time utilization, so the BEV-to-port ratio proxies utilization pressure. *Direction of bias: undetermined.* High ratios may reflect genuine under-deployment or low utilization despite available ports. The Cary, NC study [10] is limited counter-evidence not generalizing to the top-10 scope. *Future work: acquire or simulate session-level data.*

9.7 Area-Weighted Interpolation Assumes Uniform Within-Tract Population

The tract-to-ZCTA crosswalk (Section 4.5) uses area-weighted interpolation (the EPA EJScreen and HUD USPS federal standard), but weaker in large rural tracts; the validation suite (23 of 23 checks passed) confirms per-ZCTA area conservation within 1%. *Direction of bias: bounded; the urban/suburban study area is where the method is most accurate. Future work:* dasymetric interpolation; spatial sensitivity analysis.

9.8 CEJST Methodology Lock at v2.0

Justice40 designations are anchored on CEJST v2.0 (December 2024) as it stood on January 21, 2025, before the tool was removed from screeningtool.geoplatform.gov following the rescission of Executive Order 14008 on January 22, 2025. Continuity is preserved via the EDGI / Public Environmental Data Partners archive (Section 5.5) and the CEJST v2.0 Technical Support Document. *Direction of bias: unknown, depends on successor methodology. Future work:* re-run the equity pillar against any successor instrument.

9.9 NEVI Program Rule Evolution Risk

The framework is anchored on the NEVI rule (FHWA Final Rule, February 2023; Sections 3.4 and 4.5). Revisions to station spacing, minimum port count, the 150 kW DCFC threshold, or community-charging eligibility would shift the cost-effectiveness pillar inputs. *Direction of bias: unknown, rule-dependent. A loosened rule reduces cross-county discrimination, a tightened one increases it. Future work:* monitor FHWA rulemaking and rerun on material change.

9.10 State-Administered Formula Program Variance

NEVI is federally funded but state-administered, and NCDOT deployments may diverge from the scoring for legitimate non-equity reasons (right-of-way, utility coordination, supply-chain timing). NCDOT's February 2026 shift toward 16 rural and community locations (Section 1.2) is the kind of decision the framework does not predict. *Direction of bias: no statistical bias; framing risk only.* A divergence does not falsify the framework but identifies the factors NCDOT's process integrates. *Future work:* Phase 7 NCDOT deployment validation (Section 11).

9.11 Synthesis

No limitation in §§9.1–9.10 directly undermines the headline rankings in Section 6: four reinforce them (§§9.1–9.4), two are bounded (§§9.5, 9.7), one is undetermined (§9.6), and three are federal-policy or program risks (§§9.8–9.10).

10. CONCLUSION

10.1 Findings in Brief

This study delivered a defensible county-level NEVI Priority Score for the top 10 North Carolina counties by BEV registration count, led by Union (0.561), Mecklenburg (0.548), and Guilford (0.465), with the top-3 set stable across all five tested equity-weight scenarios. Out-of-sample forecast validation produced a mean absolute percentage error of 4.34% across 400 county-month observations. A Chow structural-break test identified the August 2022 Inflation Reduction Act passage as a strongly significant break ($F = 1,268.35$, $p < 1 \times 10^{-6}$), reframing the 69.00% systematic underprediction as the statistical fingerprint of accelerated post-IRA adoption rather than model failure. The three pillars are statistically independent (maximum Variance Inflation Factor of 1.41 against a conventional 5.0 threshold). A Theil-T

decomposition revealed that 84.5% of charging infrastructure inequality across the top-10 counties is *within* counties rather than between them, motivating the two-tier architecture and constituting the first structural decomposition of Theil applied to EV charging infrastructure. A spatial overlay found that 24.5% of stations within the top-10 scope sit in CEJST-designated Justice40 disadvantaged tracts, approximately proportional to the 18.5% population-weighted disadvantaged-tract share.

10.2 Summary of Findings

The five analytical phases (Section 4.1) integrate into a single prescriptive output. **Phase 1** validated SAS Model Studio BEV forecasts against a four-month out-of-sample holdout, identifying the IRA break as the explanation for systematic underprediction. **Phase 2** upgraded the supply-side baseline from a DCFC-only July 2024 extract (355 stations, 1,725 connectors) to a full February 2026 AFDC API pull (1,985 stations, 6,145 connectors). **Phase 3** measured ZIP-level inequality at both Gini-scalar (statewide population-weighted 0.566) and Theil-T levels (84.5% within-county). **Phase 4** layered workplace charging demand from LEHD LODES 2021 commuter flows through a three-layer pipeline (SE03 income filter, ACS correction at the \$75,000 threshold, 0.85 remote-work multiplier), yielding an adjusted estimate of 859,260 workers. **Phase 5** integrated CEJST v2.0 Justice40 designations into the equity pillar via an area-weighted tract-to-ZCTA crosswalk. The **NEVI scoring framework** combined these into a composite Priority Score (weights 0.40 Equity / 0.35 Utilization / 0.25 Cost-Effectiveness).

10.3 Five Unique Contributions

This study makes five contributions to the EV infrastructure allocation literature.

Contribution 1: First additive-decomposition application of the Theil index to EV charging infrastructure inequality, integrated into an allocation framework. Where prior work (notably Choi, Xu, & Jiao, 2025) uses Theil as a scalar, this study exploits the additive decomposability of the GE(1) class [2, 14]. The decomposition partitions inequality into between- and within-county components used as a design input (Sections 3.3, 4.4).

Contribution 2: First application of the HUD USPS / area-weighted crosswalk methodology to integrate CEJST Justice40 designations with AFDC station data at ZCTA resolution in North Carolina. The contribution is the integration target: overlaying CEJST v2.0 designations onto AFDC station inventory at the sub-county resolution NEVI prioritization requires (Section 4.5).

Contribution 3: NEVI-specific three-pillar scoring formulation with documented VIF orthogonality and triply-robust top-3 stability under weight perturbation. A Variance Inflation Factor maximum of 1.41 confirms pillar independence, and the top-3 ranking (Union, Mecklenburg, Guilford) holds across three perturbations (Section 4.7.1).

Contribution 4: Empirical validation that within-county inequality dominates between-county inequality in NC charging infrastructure (84.5% Theil). The 84.5% within-county share, verified to machine precision and confirmed by a Theil-L (GE(0)) check at 82.5%, justifies the two-tier architecture (Sections 3.3, 4.4, 6.3).

Contribution 5: Climate sensitivity analysis demonstrating CEJST robustness for the NC context. Removing CEJST's only forward-looking category (climate change) left 7 of 10 study counties unaffected, the rest shifting by 2.2 to 4.7 percentage points without altering the top-3 ranking (Section 4.5).

10.4 Closing Synthesis

Federal infrastructure programs increasingly require multi-criteria prioritization frameworks that reconcile efficiency with equity in a verifiable way. The NEVI Priority Score formulation makes findings actionable while remaining decision-support, not decision-making: NCDOT's actual deployment decisions will and should incorporate factors it does not model (right-of-way, utility coordination, supply-chain timing, political prioritization). What it supplies is the analytical floor a state's NEVI allocation can stand on, one that is data-driven, policy-aligned, and reproducible from the public analysis repository.

11. FUTURE WORK

The Section 9 limitations each carry a future-work pointer, consolidated here across five themes.

11.1 Scope and Data Extensions

The most immediate extension is statewide application to all 100 NC counties. Every pipeline (NCDOT BEV registrations, AFDC infrastructure, LEHD LODES commuter flows, CEJST tract designations, ACS demographics) already processes all 100 counties; only the scoring layer is scoped to the top 10 (Section 9.1). A statewide run activates the dormant `zero_station_pct` and per-county `forecast_buffer` (replacing the global 0.045; Section 9.5), raising resolution from seven to nine of nine sub-metrics and cross-validating the top-3 ranking (Section 6.6). ZIP-code-level BEV data (NCDOT or licensed IHS Markit / Experian VIO, \$5,000–\$20,000 per state) would enable ZIP-level BEV-to-port ratios, removing the two-scale mismatch (Section 5.6); LODES auxiliary files would close the 4.6% OD-Main/WAC gap (Section 5.4).

11.2 Validation and Verification

Phase 7 NCDOT deployment validation is the most consequential external check: NCDOT's February 2026 shift toward 16 rural and community NEVI locations (Section 1.2) will yield deployment data to test the output (Section 9.10). Extending the four-month holdout (July–October 2025; $n = 400$) to twelve months (July 2025–June 2026) would enable a formal Chow test of the May 2025 methodology change, currently degenerate ($n_2 = k = 2$; Section 4.9). Propagating forecast, LODES, Gini, and CEJST uncertainty into composite-score confidence intervals would extend the "triply-robust" finding (Section 4.7.1).

11.3 Methodological Refinements

Longitudinal growth-trajectory modeling, calibrated on the post-IRA August 2022 break, would replace static October 2025 BEV counts (Section 9.4). Dasymetric population-weighted interpolation (LandScan, WorldPop) would refine the area-weighted equity pillar (within 1%; Sections 4.5, 4.9, 9.7). County-specific remote-work rates would replace the uniform 0.85 multiplier (Sections 4.6, 9.2). A full DOE pass over the weight simplex would test whether the 0.40/0.35/0.25 weights (equity 0.30–0.50; Section 4.7.1) sit near a discrimination optimum.

11.4 Federal-Policy Responsiveness

Re-analysis against any CEJST v2.0 (January 21, 2025) successor (Sections 5.5, 9.8) and monitoring of FHWA NEVI Final Rule changes (Section 9.9) would test ranking durability across federal-policy regimes.

11.5 Synthesis

These items run from immediate extensions through longer-horizon refinements to policy-responsive maintenance. None is required for the Section 6 findings to stand; each strengthens or broadens the framework.

12. DATA SOURCES AND DISCLOSURES

[Personal acknowledgements removed for double-blind review.]

The study relies on publicly available data from several federal and state programs: the North Carolina Department of Transportation (Climate Change Initiatives: monthly BEV registration statistics; NEVI program announcements); the U.S. Department of Energy and the National Renewable Energy Laboratory (Alternative Fuels Data Center charging-station inventory and API access); the U.S. Census Bureau (American Community Survey 2022 five-year estimates, TIGER geographic products, and the Longitudinal Employer-Household Dynamics program for LEHD LODS origin-destination commuter data); the Council on Environmental Quality, Executive Office of the President (the original CEJST v2.0 methodology and Technical Support Document); and the Environmental Data and Governance Initiative (EDGI) together with the Public Environmental Data Partners coalition (PEDP) for archival preservation of CEJST v2.0 following the decommissioning of the federal screening tool on January 22, 2025.

Funding. This study received no external funding. It is self-directed research with no corporate sponsor.

Data and Code Availability. All analytical scripts, processed data outputs, raw datasets, figure-generation code, and validation suites used to produce this manuscript are publicly available in an open repository under a noncommercial license. The repository URL is withheld here for double-blind review and will be provided in the camera-ready version. A provenance table documents the source and pull date for each dataset, and the analysis is reproducible from a clean clone using the committed dependency lockfile (uv.lock) and the published scripts against the committed raw data.

Conflicts of Interest. The author declares no conflicts of interest.

Generative AI Disclosure. Generative AI tools were used for code generation and refactoring, structured role-play methodology review, literature-review filtering, and prose drafting and polishing. All AI outputs were verified against source data through a Claims-to-Evidence Matrix; AI tools are research and writing aids, not co-authors, and the author retains sole responsibility for the manuscript and its conclusions.

APPENDICES

APPENDIX A: DATA QUALITY METHODOLOGY REVIEW (SUMMARY)

This appendix summarizes an internal data quality methodology review of the analytical pipeline, conducted in April 2026 against a fixed repository commit using a structured AI-assisted role-play methodology.

A.1 Scope and Outcome

The review covered approximately sixty distinct items spanning the six standard data quality dimensions (Accuracy, Completeness, Consistency, Timeliness, Validity, Uniqueness) and a ten-section data quality roadmap. Of the roughly sixty items, zero remained open at evaluation snapshot; every item resolved to a strength, a limitation with documented direction of bias, or a future-work pointer, with none deferred or left ambiguous.

A.2 Panel and Selected Findings

Five specialist lanes contributed to the manuscript: data quality engineering (six-dimension audit; Section 4 framing and Section 5 governance), spatial/GIS data (Section 4.5), federal data and policy compliance (Sections 5.5, 9.8, and Section 9 bias framing), econometrics and forecast validation (Sections 4.9, 6.1, 6.6), and business and data analytics strategy (Sections 7.2, 7.3, and Section 9 weakness ordering).

Substantive findings adopted by the body include:

- **Multicollinearity.** Maximum Variance Inflation Factor across the three scoring pillars is 1.41, well below the conventional 5.0 threshold (Manuscript §4.9 and §6.6).
- **CEJST archival pathway.** The federal CEJST tool was decommissioned on January 22, 2025; the EDGI / Public Environmental Data Partners archive is the operational source for v2.0 (Manuscript §5.5 and §9.8).
- **Federal spatial-standard alignment.** The spatial validation suite passed at 23-of-23 (Manuscript §4.5).
- **Direction-of-bias disclosure pattern** anchors Manuscript §9.
- **Three-archetype framing** appears in Manuscript §7.3.
- **Data Governance Readiness chain.** The 8-point lineage chain is summarized in Manuscript §5.6.

The complete review, a self-contained companion document (withheld for double-blind review), details the roadmap walkthrough, the six-dimension audit, the Data Cleaning Log, and the risk-assessment statement condensed into Manuscript §9.

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